

DE CA-2

Completed answers and dashboard-backed analysis using the provided BigMart Sales.csv

Data note: The supplied CSV has 5,681 rows and 11 fields, but it does not include actual Sales, Profit, transaction Date/Month, or a named Region column. Therefore, the dashboard below uses available substitutes: **Item_MRP** as the value measure, **Item_Type** as category, **Outlet_Location_Type** as region, and **Outlet_Establishment_Year** as the time field. True monthly sales trend, seasonality, and profit/loss conclusions require those missing fields.

Metric	Value
Rows in dataset	5,681
Value measure used	Item_MRP
Highest average MRP category	Starchy Foods (152.09)
Lowest average MRP category	Baking Goods (129.05)
Highest total MRP location tier	Tier 3 (315,438)
Lowest average MRP location tier	Tier 2 (140.72)

Part A: Dashboard

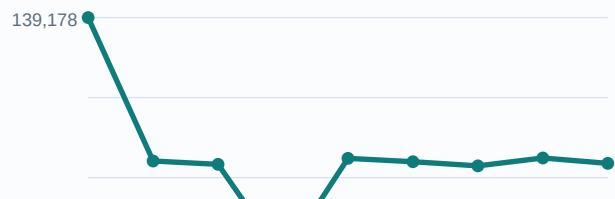
Dashboard Snapshot - BigMart Sales CSV

Region: All

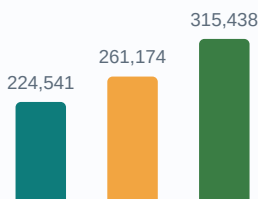
Category: All

Year: All

MRP Trend by Outlet Establishment Year



MRP by Region



Average MRP by Category



Part A: Answers

Q1. Identify Trend

Using the available time field, Outlet_Establishment_Year, the value measure fluctuates rather than showing a steady increase or decrease. Total Item_MRP is highest for outlets established in 1985, drops in some later establishment years, and rises again around 2007. The average MRP by year is also fairly stable, mostly between about 138 and 143, so the visual does not show a strong upward sales trend.

Q2. Identify Seasonality

Monthly seasonality cannot be reliably identified from the supplied BigMart CSV because there is no transaction date, month, or monthly sales column. A correct seasonality conclusion would need monthly sales values over multiple years. With only outlet establishment year, there is no evidence for consistent high or low months.

Q3. Business Insight

Actual profit and loss cannot be calculated because the CSV does not contain Profit, Cost, Quantity, or Sales fields. Using average Item_MRP only as a weak proxy, Baking Goods has the lowest average MRP (129.05), while Tier 2 has the slightly lowest average MRP by location (140.72). Recommendation: collect actual sales and profit/margin data, then review pricing, promotions, and assortment for lower-performing categories before calling any segment loss-making.

Lowest average MRP categories

Category	Average MRP	Total MRP	Rows
Baking Goods	129.05	56,525	438
Breakfast	132.48	10,069	76
Frozen Foods	133.41	76,044	570
Others	135.11	14,997	111
Health and Hygiene	136.63	46,181	338
Canned	136.92	59,558	435

Part B: Time-Series Forecasting

Q4. Why is stationarity important before applying ARIMA?

Stationarity is important because ARIMA works best when the statistical properties of the series, such as mean, variance, and autocorrelation, remain stable over time. If the data has trend or seasonality, the model may learn misleading relationships and produce unreliable forecasts. Differencing is commonly used to make a non-stationary series more stationary before fitting ARIMA.

Q5. Difference between ARIMA and SARIMA

ARIMA models non-seasonal time-series behavior using autoregressive, differencing, and moving average terms: $ARIMA(p, d, q)$. SARIMA extends ARIMA by adding seasonal terms: $SARIMA(p, d, q)(P, D, Q, s)$. SARIMA is preferred when the series has a repeating seasonal cycle, such as weekly demand patterns, monthly yearly cycles, or annual holiday peaks.

Q6. Case study: E-commerce sales forecasting

SARIMA should be chosen for this scenario. The sales series has an increasing trend and a repeating seasonal pattern where sales rise sharply every November and December, then fall after December and repeat the same way each year. ARIMA can handle trend after differencing, but it does not explicitly model the repeating yearly seasonal effect. SARIMA is better because it includes both non-seasonal and seasonal components. The 'S' in SARIMA stands for Seasonal; for monthly sales data, the seasonal period would typically be 12 months.

Scenario feature	Model implication
Steady increasing trend	Use differencing to stabilize the series.
Repeating November-December spike	Use seasonal terms with period 12.
Need 6-month forecast	Fit SARIMA and forecast the next six monthly points.
Inventory planning	Use forecasts plus safety stock before the festive months.